

From Maxwell to the Electromagnetic Field Tensor

A Logical Construction of the Covariant Structure of Classical Electromagnetism

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1 Purpose and logical strategy

The standard differential form of Maxwell's equations in SI units is

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}, \quad (1)$$

$$\nabla \cdot \mathbf{B} = 0, \quad (2)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad (3)$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \quad (4)$$

These equations are fully compatible with special relativity, but this compatibility is not manifest in their usual three-dimensional form. The central goal of this document is to explain, step by step, why the relativistic formulation of electromagnetism naturally introduces

1. the four-current

$$J^\mu = (c\rho, \mathbf{J}),$$

2. the antisymmetric electromagnetic tensor

$$F_{\mu\nu},$$

3. the covariant Maxwell equations

$$\partial_\mu F^{\mu\nu} = -\mu_0 J^\nu, \quad \partial_{[\alpha} F_{\beta\gamma]} = 0.$$

The important point is that these objects should not be regarded as clever notational packages invented after the fact. Their structure follows from the physical requirements of special relativity, charge conservation, and the relativistic form of the Lorentz force.

Throughout the document we use the convention

$$x^\mu = (ct, x, y, z), \quad \eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1),$$

so that

$$\partial_\mu = \left(\frac{1}{c} \frac{\partial}{\partial t}, \nabla \right).$$

2 Why do we want a covariant formulation?

2.1 The principle of special relativity

The principle of special relativity states that the laws of physics must have the same form in all inertial reference frames. Since inertial frames are related by Lorentz transformations, fundamental equations should be Lorentz¹ covariant.

If two inertial frames S and S' are related by a Lorentz transformation,

$$x'^{\mu} = \Lambda^{\mu}_{\nu} x^{\nu}, \quad (5)$$

then a fundamental law should retain the same geometrical form in both frames.

This requirement motivates the use of tensors.

2.2 Tensorial covariance

A contravariant four-vector transforms as

$$V'^{\mu} = \Lambda^{\mu}_{\nu} V^{\nu}. \quad (6)$$

A rank-two contravariant tensor transforms as

$$T'^{\mu\nu} = \Lambda^{\mu}_{\alpha} \Lambda^{\nu}_{\beta} T^{\alpha\beta}. \quad (7)$$

Suppose a physical law is written as

$$A^{\mu} = B^{\mu}. \quad (8)$$

Then, under a Lorentz transformation,

$$A'^{\mu} = \Lambda^{\mu}_{\nu} A^{\nu} = \Lambda^{\mu}_{\nu} B^{\nu} = B'^{\mu}. \quad (9)$$

Thus an equality between tensors remains valid in every Lorentz-related inertial frame.

This is the precise connection between the word *covariant* and the tensor transformation law:

A manifestly covariant equation is written entirely in terms of geometrical objects whose transformation laws guarantee that the equation keeps the same form under Lorentz transformations.

The usual three-dimensional Maxwell equations are Lorentz covariant, but this fact is hidden because they separately refer to time, space, electric field, magnetic field, charge density, and current density. The relativistic formulation reorganizes these quantities into genuine spacetime objects.

¹Being more precise, in a Minkowskian space-time two inertial frame are connected by elements of the the Poincaré Group, therefore we should talk of Poincaré covariance.

3 The first relativistic unification: charge and current

3.1 From charge density to current density

Consider a distribution of charged matter. In a generic inertial frame its charge density is ρ and its current density is

$$\mathbf{J} = \rho \mathbf{u}, \quad (10)$$

where $\mathbf{u}$ is the velocity of the charged matter.

Let ρ_0 denote the proper charge density, namely the density measured in the rest frame of the charges.

The four-velocity of the charged matter is

$$U^\mu = \gamma_u(c, \mathbf{u}), \quad \gamma_u = \frac{1}{\sqrt{1 - u^2/c^2}}. \quad (11)$$

Since ρ_0 is a Lorentz scalar, the product

$$J^\mu \equiv \rho_0 U^\mu \quad (12)$$

is automatically a four-vector.

Using the relativistic relation

$$\rho = \gamma_u \rho_0, \quad (13)$$

we obtain

$$J^\mu = \rho_0 \gamma_u(c, \mathbf{u}) \quad (14)$$

$$= (c\rho, \rho \mathbf{u}). \quad (15)$$

Since $\mathbf{J} = \rho \mathbf{u}$,

$$\boxed{J^\mu = (c\rho, \mathbf{J})}. \quad (16)$$

Thus charge density and current density are not independent relativistic quantities. They are the temporal and spatial components of one spacetime vector.

3.2 Explicit Lorentz transformation of charge and current

Consider a boost with velocity v along the x -axis:

$$\Lambda^\mu{}_\nu = \begin{pmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \beta = \frac{v}{c}, \quad \gamma = \frac{1}{\sqrt{1 - \beta^2}}. \quad (17)$$

Using

$$J'^\mu = \Lambda^\mu{}_\nu J^\nu$$

with

$$J^\mu = \begin{pmatrix} c\rho \\ J_x \\ J_y \\ J_z \end{pmatrix},$$

we obtain

$$c\rho' = \gamma(c\rho - \beta J_x), \quad (18)$$

$$J'_x = \gamma(J_x - \beta c\rho), \quad (19)$$

$$J'_y = J_y, \quad (20)$$

$$J'_z = J_z. \quad (21)$$

Therefore

$$\boxed{\rho' = \gamma \left(\rho - \frac{v J_x}{c^2} \right)} \quad (22)$$

and

$$\boxed{J'_x = \gamma(J_x - v\rho), \quad J'_y = J_y, \quad J'_z = J_z.} \quad (23)$$

For a boost of arbitrary velocity $\mathbf{v}$,

$$\rho' = \gamma \left(\rho - \frac{\mathbf{v} \cdot \mathbf{J}}{c^2} \right), \quad (24)$$

$$\mathbf{J}'_{\parallel} = \gamma(\mathbf{J}_{\parallel} - \rho \mathbf{v}), \quad (25)$$

$$\mathbf{J}'_{\perp} = \mathbf{J}_{\perp}. \quad (26)$$

This is a fundamental relativistic fact: charge density and current density mix under Lorentz transformations.

3.3 Physical example

Suppose a wire is electrically neutral in S ,

$$\rho = 0,$$

but carries a current along x ,

$$J_x \neq 0.$$

An observer moving parallel to the wire measures

$$\rho' = -\gamma \frac{v J_x}{c^2}.$$

Thus a configuration that is purely a current for one observer can possess a nonzero charge density for another observer. This already suggests that electric and magnetic phenomena must be related by Lorentz transformations.

4 Why the electromagnetic field must be a tensor

4.1 Relativistic form of the Lorentz force

The ordinary Lorentz force is

$$\mathbf{f} = q(\mathbf{E} + \mathbf{u} \times \mathbf{B}). \quad (27)$$

In relativity, the momentum of a massive particle is the four-vector

$$p^\mu = mU^\mu. \quad (28)$$

The four-force is defined by

$$f^\mu = \frac{dp^\mu}{d\tau}. \quad (29)$$

Since p^μ is a four-vector and τ is a Lorentz scalar, f^μ is also a four-vector.

Using

$$p^\mu = \left(\frac{\mathcal{E}}{c}, \mathbf{p} \right),$$

one finds

$$f^\mu = \gamma_u \left(\frac{\mathbf{f} \cdot \mathbf{u}}{c}, \mathbf{f} \right). \quad (30)$$

The electromagnetic force law must therefore be expressible as a four-vector equation.

4.2 Assumptions entering the relativistic force law

We now impose the following physically motivated requirements:

- (i) The electromagnetic force is local²;
- (ii) It is proportional to the electric charge q ;
- (iii) It depends on the electromagnetic field evaluated at the particle position;
- (iv) It is Lorentz covariant;
- (v) It is linear in the particle four-velocity U^μ , consistently with the observed Lorentz force law.

A linear map that sends the four-vector U^ν into another four-vector must be represented by a rank-two tensor³

. We therefore write

$$f^\mu = qF^\mu{}_\nu U^\nu. \quad (31)$$

The transformation law of $F^\mu{}_\nu$ is required in order for (31) to retain the same form in all inertial frames.

5 Why the electromagnetic tensor is antisymmetric

For a particle of fixed rest mass,

$$U^\mu U_\mu = -c^2. \quad (32)$$

Differentiating with respect to proper time,

$$\frac{d}{d\tau}(U^\mu U_\mu) = 2U_\mu \frac{dU^\mu}{d\tau} = 0. \quad (33)$$

Multiplying by m ,

$$U_\mu f^\mu = 0. \quad (34)$$

Using (31),

$$qU_\mu F^\mu{}_\nu U^\nu = 0, \quad (35)$$

or

$$F_{\mu\nu} U^\mu U^\nu = 0. \quad (36)$$

Now decompose $F_{\mu\nu}$ into symmetric and antisymmetric parts:

$$F_{\mu\nu} = F_{(\mu\nu)} + F_{[\mu\nu]}. \quad (37)$$

²In this case the word local is used to underline the concept that the force in a particular event depends only upon physical quantities defined in that same event

³Algebraically, $F^\mu{}_\nu$ represents a linear map $F : V \rightarrow V$. Given a basis $\{e_\mu\}$ and a vector $U = U^\nu e_\nu$, linearity gives $F(U) = U^\nu F(e_\nu)$. Since $F(e_\nu) \in V$, one may write $F(e_\nu) = F^\mu{}_\nu e_\mu$, and therefore $F(U) = F^\mu{}_\nu U^\nu e_\mu$, so that $[F(U)]^\mu = F^\mu{}_\nu U^\nu$. Hence $F \in \text{Hom}(V, V) \simeq V \otimes V^*$, i.e. F is naturally a tensor of type $(1, 1)$.

Because

$$U^\mu U^\nu$$

is symmetric⁴ we get that,

$$F_{[\mu\nu]}U^\mu U^\nu = 0$$

Hence (36) for arbitrary allowed four-velocities requires

$$F_{(\mu\nu)} = 0.$$

Therefore

$$\boxed{F_{\mu\nu} = -F_{\nu\mu}} \quad (38)$$

This is the fundamental reason for the antisymmetry of the electromagnetic tensor in the relativistic force formulation.

A 4×4 antisymmetric tensor⁵ has

$$\frac{4(4-1)}{2} = 6$$

independent components. This matches the six components contained in the pair $(\mathbf{E}, \mathbf{B})$, but this counting is a consistency check rather than the original motivation.

6 Identifying the six components with $\mathbf{E}$ and $\mathbf{B}$

The four-velocity is

$$U^\mu = \gamma_u(c, u_x, u_y, u_z). \quad (39)$$

The spatial components of the four-force satisfy

$$f^i = \gamma_u f_{3D}^i. \quad (40)$$

Using the ordinary Lorentz force,

$$f^i = q\gamma_u [E_i + (\mathbf{u} \times \mathbf{B})_i]. \quad (41)$$

From the covariant force law,

$$f^i = qF^i{}_\nu U^\nu \quad (42)$$

$$= q\gamma_u (cF^i{}_0 + F^i{}_j u_j). \quad (43)$$

Comparing (41) and (43) yields

$$F^i{}_0 = \frac{E_i}{c}, \quad (44)$$

and

$$F^i{}_j u_j = (\mathbf{u} \times \mathbf{B})_i = \varepsilon_{ijk} u_j B_k. \quad (45)$$

Thus

$$F^i{}_j = \varepsilon_{ijk} B_k. \quad (46)$$

⁴The contraction of an antisymmetric tensor with the symmetric product $U^\mu U^\nu$ vanishes identically. Indeed, $F_{[\mu\nu]}U^\mu U^\nu = F_{[\nu\mu]}U^\nu U^\mu = -F_{[\mu\nu]}U^\mu U^\nu$, where we used $F_{[\nu\mu]} = -F_{[\mu\nu]}$ and $U^\nu U^\mu = U^\mu U^\nu$.

Hence $F_{[\mu\nu]}U^\mu U^\nu = 0$.

⁵The general equation to calculate the independent components of a $N \times N$ antisymmetric tensor is: $\frac{n(n-1)}{2}$

With the metric convention $(-+++)$, one obtains

$$\boxed{F^{0i} = \frac{E_i}{c}, \quad F^{i0} = -\frac{E_i}{c}, \quad F^{ij} = \varepsilon_{ijk} B_k.} \quad (47)$$

Hence

$$\boxed{F^{\mu\nu} = \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & B_z & -B_y \\ -E_y/c & -B_z & 0 & B_x \\ -E_z/c & B_y & -B_x & 0 \end{pmatrix}.} \quad (48)$$

7 How electric and magnetic fields mix under Lorentz transformations

Because $F^{\mu\nu}$ is a tensor,

$$\boxed{F'^{\mu\nu} = \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta F^{\alpha\beta}.} \quad (49)$$

For a boost along x , the Lorentz matrix mixes the 0 and 1 indices. Since

$$E_y \sim F^{02}, \quad B_z \sim F^{12},$$

the boost necessarily mixes E_y and B_z .

For example,

$$F'^{02} = \Lambda^0{}_\alpha \Lambda^2{}_\beta F^{\alpha\beta} \quad (50)$$

$$= \Lambda^0{}_0 F^{02} + \Lambda^0{}_1 F^{12} \quad (51)$$

$$= \gamma \left(\frac{E_y}{c} \right) - \gamma \beta B_z \quad (52)$$

$$= \gamma \frac{E_y}{c} - \gamma \frac{v}{c} B_z. \quad (53)$$

Since

$$F'^{02} = \frac{E'_y}{c},$$

we obtain

$$\boxed{E'_y = \gamma(E_y - v B_z).} \quad (54)$$

Repeating the calculation for all components gives

$$E'_x = E_x, \quad (55)$$

$$E'_y = \gamma(E_y - v B_z), \quad (56)$$

$$E'_z = \gamma(E_z + v B_y), \quad (57)$$

and

$$B'_x = B_x, \quad (58)$$

$$B'_y = \gamma \left(B_y + \frac{v}{c^2} E_z \right), \quad (59)$$

$$B'_z = \gamma \left(B_z - \frac{v}{c^2} E_y \right). \quad (60)$$

In vector form,

$$\boxed{\mathbf{E}'_{\parallel} = \mathbf{E}_{\parallel}, \quad \mathbf{B}'_{\parallel} = \mathbf{B}_{\parallel},} \quad (61)$$

while

$$\boxed{\mathbf{E}'_{\perp} = \gamma(\mathbf{E}_{\perp} + \mathbf{v} \times \mathbf{B}),} \quad (62)$$

and

$$\boxed{\mathbf{B}'_{\perp} = \gamma \left(\mathbf{B}_{\perp} - \frac{1}{c^2} \mathbf{v} \times \mathbf{E} \right).} \quad (63)$$

Thus electric and magnetic fields are observer-dependent decompositions of one spacetime tensor.

8 A physical interpretation of the field mixing

Suppose that in one inertial frame

$$\mathbf{B} = 0, \quad \mathbf{E} \neq 0.$$

For an observer moving with velocity $\mathbf{v}$,

$$\mathbf{B}'_{\perp} = -\gamma \frac{\mathbf{v} \times \mathbf{E}}{c^2}, \quad (64)$$

which is generally nonzero.

Likewise, if

$$\mathbf{E} = 0, \quad \mathbf{B} \neq 0,$$

then

$$\mathbf{E}'_{\perp} = \gamma \mathbf{v} \times \mathbf{B}. \quad (65)$$

Therefore “electric field” and “magnetic field” are not independent relativistic objects. They are frame-dependent parts of $F_{\mu\nu}$.

The same structural phenomenon occurred for charge and current:

$$(\rho, \mathbf{J}) \longrightarrow J^{\mu}.$$

We therefore have two distinct relativistic unifications:

$$\boxed{(\rho, \mathbf{J}) \longrightarrow J^{\mu},} \quad (66)$$

and

$$\boxed{(\mathbf{E}, \mathbf{B}) \longrightarrow F_{\mu\nu}.} \quad (67)$$

9 The homogeneous Maxwell equations and the four-potential

In the first part of this paragraph we are going to show that the equations

$$\begin{aligned} \nabla \cdot \mathbf{B} &= 0 \\ \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} &= 0 \end{aligned}$$

can be rewritten as

$$\boxed{\partial_{[\alpha} F_{\beta\gamma]} = 0.} \quad (68)$$

Equivalently,

$$\partial_{\alpha} F_{\beta\gamma} + \partial_{\beta} F_{\gamma\alpha} + \partial_{\gamma} F_{\alpha\beta} = 0. \quad (69)$$

In order to do so, let us define

$$H_{\alpha\beta\gamma} \equiv \partial_{\alpha} F_{\beta\gamma} + \partial_{\beta} F_{\gamma\alpha} + \partial_{\gamma} F_{\alpha\beta}.$$

Since $F_{\mu\nu} = -F_{\nu\mu}$, the tensor $H_{\alpha\beta\gamma}$ is completely antisymmetric. Therefore, in four spacetime dimensions, it has only

$$\binom{4}{3} = 4$$

independent components.

Using

$$F_{0i} = -\frac{E_i}{c}, \quad F_{i0} = +\frac{E_i}{c}, \quad F_{ij} = \epsilon_{ijk}B_k, \quad \partial_0 = \frac{1}{c}\partial_t,$$

the purely spatial component gives

$$\begin{aligned} H_{123} &= \partial_1 F_{23} + \partial_2 F_{31} + \partial_3 F_{12} \\ &= \partial_x B_x + \partial_y B_y + \partial_z B_z = \nabla \cdot \mathbf{B}. \end{aligned} \quad (70)$$

Hence

$$H_{123} = 0 \quad \Longleftrightarrow \quad \nabla \cdot \mathbf{B} = 0.$$

The remaining three independent components contain one temporal index. For example,

$$\begin{aligned} H_{012} &= \partial_0 F_{12} + \partial_1 F_{20} + \partial_2 F_{01} \\ &= \frac{1}{c} \left(\frac{\partial B_z}{\partial t} + \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \right) \\ &= \frac{1}{c} \left[\frac{\partial B_z}{\partial t} + (\nabla \times \mathbf{E})_z \right]. \end{aligned} \quad (71)$$

Similarly,

$$H_{023} = \frac{1}{c} \left[\frac{\partial B_x}{\partial t} + (\nabla \times \mathbf{E})_x \right],$$

and

$$H_{031} = \frac{1}{c} \left[\frac{\partial B_y}{\partial t} + (\nabla \times \mathbf{E})_y \right].$$

Therefore

$$H_{0ij} = 0 \quad \Longleftrightarrow \quad \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = 0.$$

Thus the four independent equations contained in $H_{\alpha\beta\gamma} = 0$ are precisely the magnetic Gauss law and the three components of Faraday's law. Conversely, if these two Maxwell equations hold, all four independent components of $H_{\alpha\beta\gamma}$ vanish; since $H_{\alpha\beta\gamma}$ is completely antisymmetric, every other component either vanishes because two indices coincide or differs from one of these four only by a sign. Consequently,

$$\boxed{\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = 0 \quad \Longleftrightarrow \quad \partial_\alpha F_{\beta\gamma} + \partial_\beta F_{\gamma\alpha} + \partial_\gamma F_{\alpha\beta} = 0.}$$

Equivalently,

$$\boxed{\partial_{[\alpha} F_{\beta\gamma]} = 0.}$$

Before logically deriving the four-potential, we should observe that in the language of differential forms, the electromagnetic field is the two-form⁶

$$F = \frac{1}{2} F_{\mu\nu} dx^\mu \wedge dx^\nu, \quad (76)$$

Taking the exterior derivative gives

$$dF = \frac{1}{2} d(F_{\mu\nu} dx^\mu \wedge dx^\nu) = \frac{1}{2} dF_{\mu\nu} \wedge dx^\mu \wedge dx^\nu, \quad (77)$$

since

$$d(dx^\mu) = 0.$$

Because the components $F_{\mu\nu}$ are scalar functions of the coordinates,

$$dF_{\mu\nu} = \partial_\alpha F_{\mu\nu} dx^\alpha,$$

and therefore

$$dF = \frac{1}{2} \partial_\alpha F_{\mu\nu} dx^\alpha \wedge dx^\mu \wedge dx^\nu. \quad (78)$$

The three-form

$$dx^\alpha \wedge dx^\mu \wedge dx^\nu$$

is completely antisymmetric in its indices. Hence only the completely antisymmetric part of $\partial_\alpha F_{\mu\nu}$ contributes. Using the antisymmetry $F_{\mu\nu} = -F_{\nu\mu}$, one obtains

$$dF = \frac{1}{3!} (\partial_\alpha F_{\mu\nu} + \partial_\mu F_{\nu\alpha} + \partial_\nu F_{\alpha\mu}) dx^\alpha \wedge dx^\mu \wedge dx^\nu. \quad (79)$$

⁶For an antisymmetric rank-two tensor,

$$F_{\mu\nu} = -F_{\nu\mu},$$

the ordinary tensor-product representation is

$$F = F_{\mu\nu} dx^\mu \otimes dx^\nu.$$

Using the antisymmetry of $F_{\mu\nu}$, we may write

$$F = \frac{1}{2} F_{\mu\nu} dx^\mu \otimes dx^\nu + \frac{1}{2} F_{\mu\nu} dx^\mu \otimes dx^\nu. \quad (72)$$

In the second term, exchanging the dummy indices $\mu \leftrightarrow \nu$ gives

$$F = \frac{1}{2} F_{\mu\nu} dx^\mu \otimes dx^\nu + \frac{1}{2} F_{\nu\mu} dx^\nu \otimes dx^\mu. \quad (73)$$

Since

$$F_{\nu\mu} = -F_{\mu\nu},$$

it follows that

$$F = \frac{1}{2} F_{\mu\nu} (dx^\mu \otimes dx^\nu - dx^\nu \otimes dx^\mu). \quad (74)$$

By definition of the exterior product,

$$dx^\mu \wedge dx^\nu \equiv dx^\mu \otimes dx^\nu - dx^\nu \otimes dx^\mu,$$

and therefore

$$\boxed{F = \frac{1}{2} F_{\mu\nu} dx^\mu \wedge dx^\nu.} \quad (75)$$

Hence, for an antisymmetric tensor,

$$\boxed{F_{\mu\nu} dx^\mu \otimes dx^\nu = \frac{1}{2} F_{\mu\nu} dx^\mu \wedge dx^\nu.}$$

Since the independent basis three-forms $dx^\alpha \wedge dx^\mu \wedge dx^\nu$ are linearly independent, dF vanishes if and only if each corresponding coefficient vanishes. Therefore

$$\boxed{dF = 0 \iff \partial_\alpha F_{\mu\nu} + \partial_\mu F_{\nu\alpha} + \partial_\nu F_{\alpha\mu} = 0} \quad (80)$$

or, equivalently,

$$\boxed{\partial_{[\alpha} F_{\mu\nu]} = 0.} \quad (81)$$

Thus the homogeneous Maxwell equations state that the electromagnetic two-form F is closed. By the Poincaré lemma, every closed form is locally exact, so that, at least locally, there exists a one-form

$$A = A_\mu dx^\mu \quad (82)$$

such that

$$F = dA. \quad (83)$$

The component of this relation can be obtained explicitly. Starting from

$$\begin{aligned} dA &= d(A_\nu dx^\nu) \\ &= dA_\nu \wedge dx^\nu \\ &= \partial_\mu A_\nu dx^\mu \wedge dx^\nu, \end{aligned} \quad (84)$$

we exploit the antisymmetry of the wedge product. Writing the same expression with the dummy indices exchanged,

$$\partial_\nu A_\mu dx^\nu \wedge dx^\mu = -\partial_\nu A_\mu dx^\mu \wedge dx^\nu,$$

and averaging the two equivalent expressions gives

$$dA = \frac{1}{2} (\partial_\mu A_\nu - \partial_\nu A_\mu) dx^\mu \wedge dx^\nu. \quad (85)$$

On the other hand,

$$F = \frac{1}{2} F_{\mu\nu} dx^\mu \wedge dx^\nu.$$

Since $F = dA$ and the basis two-forms $dx^\mu \wedge dx^\nu$ are linearly independent, their coefficients must coincide. Hence

$$\boxed{F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.} \quad (86)$$

Thus the usual component expression for the electromagnetic tensor is not introduced independently: it follows locally from the homogeneous Maxwell equations through

$$\partial_{[\alpha} F_{\mu\nu]} = 0 \iff dF = 0 \implies F = dA.$$

This logical order is important: one need not define $F_{\mu\nu}$ through A_μ at the beginning. One can first identify $F_{\mu\nu}$ through the relativistic force law and then infer the local existence of a potential from the homogeneous Maxwell equations.

10 From Gauss and Ampère-Maxwell to the four-current equation

The remaining two Maxwell equations are

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}, \quad (87)$$

$$\nabla \times \mathbf{B} - \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t} = \mu_0 \mathbf{J}. \quad (88)$$

We now show that both are contained in

$$\boxed{\partial_\mu F^{\mu\nu} = -\mu_0 J^\nu.} \quad (89)$$

10.1 The temporal component $\nu = 0$

For $\nu = 0$,

$$\partial_\mu F^{\mu 0} = -\mu_0 J^0. \quad (90)$$

Since $F^{00} = 0$,

$$\partial_i F^{i0} = -\mu_0 c \rho. \quad (91)$$

Using

$$F^{i0} = -\frac{E_i}{c},$$

we find

$$-\frac{1}{c} \partial_i E_i = -\mu_0 c \rho. \quad (92)$$

Hence

$$\nabla \cdot \mathbf{E} = \mu_0 c^2 \rho. \quad (93)$$

Since

$$c^2 = \frac{1}{\mu_0 \varepsilon_0},$$

we obtain

$$\boxed{\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}}. \quad (94)$$

Thus Gauss's law is precisely the temporal component of the covariant equation.

10.2 The spatial components $\nu = i$

For $\nu = i$,

$$\partial_\mu F^{\mu i} = -\mu_0 J^i. \quad (95)$$

Separating temporal and spatial derivatives,

$$\partial_0 F^{0i} + \partial_j F^{ji} = -\mu_0 J^i. \quad (96)$$

Now

$$F^{0i} = \frac{E_i}{c},$$

so

$$\partial_0 F^{0i} = \frac{1}{c^2} \frac{\partial E_i}{\partial t}. \quad (97)$$

Also, from

$$F^{ij} = \varepsilon_{ijk} B_k,$$

we have

$$F^{ji} = -\varepsilon_{ijk} B_k,$$

therefore

$$\partial_j F^{ji} = -\varepsilon_{ijk} \partial_j B_k = -(\nabla \times \mathbf{B})_i. \quad (98)$$

Hence

$$\frac{1}{c^2} \frac{\partial E_i}{\partial t} - (\nabla \times \mathbf{B})_i = -\mu_0 J_i. \quad (99)$$

In vector notation,

$$\boxed{\nabla \times \mathbf{B} - \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t} = \mu_0 \mathbf{J}}. \quad (100)$$

This is the Ampère-Maxwell equation.

Thus

$$\boxed{\partial_\mu F^{\mu\nu} = -\mu_0 J^\nu} \quad (101)$$

contains both Gauss's electric law and the Ampère-Maxwell law.

10.3 Inverse relation

Conversely, the electric Gauss law and the Ampère–Maxwell law imply the covariant equation. For the temporal component, using $F^{i0} = -E_i/c$ and $J^0 = c\rho$,

$$\partial_\mu F^{\mu 0} = \partial_i F^{i0} = -\frac{1}{c} \nabla \cdot \mathbf{E} = -\frac{\rho}{c\varepsilon_0} = -\mu_0 c\rho = -\mu_0 J^0. \quad (102)$$

For a spatial component $\nu = i$, using $F^{0i} = E_i/c$, $F^{ji} = -\epsilon_{ijk}B_k$, and $\partial_0 = c^{-1}\partial_t$,

$$\begin{aligned} \partial_\mu F^{\mu i} &= \partial_0 F^{0i} + \partial_j F^{ji} \\ &= \frac{1}{c^2} \frac{\partial E_i}{\partial t} - (\nabla \times \mathbf{B})_i = -\mu_0 J^i, \end{aligned} \quad (103)$$

where the last equality follows from the Ampère–Maxwell equation. Since $\nu = 0, 1, 2, 3$ exhausts all components, the two Maxwell equations imply

$$\boxed{\partial_\mu F^{\mu\nu} = -\mu_0 J^\nu.}$$

Together with the previous derivation, this establishes the equivalence

$$\boxed{\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}, \quad \nabla \times \mathbf{B} - \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t} = \mu_0 \mathbf{J} \quad \Longleftrightarrow \quad \partial_\mu F^{\mu\nu} = -\mu_0 J^\nu.}$$

11 Why $\partial_\mu F^{\mu\nu}$ is a four-vector

To see explicitly why the inhomogeneous Maxwell equation is covariant, start from

$$F'^{\mu\nu} = \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma F^{\rho\sigma}. \quad (104)$$

The derivative transforms as

$$\partial'_\mu = \Lambda_\mu{}^\alpha \partial_\alpha. \quad (105)$$

Therefore

$$\partial'_\mu F'^{\mu\nu} = \Lambda_\mu{}^\alpha \partial_\alpha (\Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma F^{\rho\sigma}). \quad (106)$$

For a global Lorentz transformation, $\Lambda^\mu{}_\nu$ is constant, so

$$\partial'_\mu F'^{\mu\nu} = \Lambda_\mu{}^\alpha \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma \partial_\alpha F^{\rho\sigma}. \quad (107)$$

Using

$$\Lambda_\mu{}^\alpha \Lambda^\mu{}_\rho = \delta^\alpha{}_\rho, \quad (108)$$

we get

$$\boxed{\partial'_\mu F'^{\mu\nu} = \Lambda^\nu{}_\sigma \partial_\rho F^{\rho\sigma}.} \quad (109)$$

Therefore

$$\partial_\mu F^{\mu\nu}$$

transforms exactly as a four-vector. Since J^ν is also a four-vector, the equation

$$\partial_\mu F^{\mu\nu} = -\mu_0 J^\nu$$

is manifestly Lorentz covariant.

12 The final covariant form of Maxwell's equations

The four standard differential Maxwell equations are therefore equivalent to the pair

$$\boxed{\partial_\mu F^{\mu\nu} = -\mu_0 J^\nu} \quad (110)$$

and

$$\boxed{\partial_{[\alpha} F_{\beta\gamma]} = 0.} \quad (111)$$

Introducing the dual tensor

$$\tilde{F}^{\mu\nu} = \frac{1}{2} \varepsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}, \quad (112)$$

the homogeneous equations may also be written as

$$\boxed{\partial_\mu \tilde{F}^{\mu\nu} = 0.} \quad (113)$$

Thus the complete Maxwell system becomes

$$\boxed{\partial_\mu F^{\mu\nu} = -\mu_0 J^\nu, \quad \partial_\mu \tilde{F}^{\mu\nu} = 0.} \quad (114)$$

13 Charge conservation as a consequence

Take the divergence of the inhomogeneous Maxwell equation:

$$\partial_\nu \partial_\mu F^{\mu\nu} = -\mu_0 \partial_\nu J^\nu. \quad (115)$$

The operator

$$\partial_\nu \partial_\mu$$

is symmetric under $\mu \leftrightarrow \nu$, whereas

$$F^{\mu\nu}$$

is antisymmetric. Their contraction therefore vanishes:

$$\partial_\nu \partial_\mu F^{\mu\nu} = 0. \quad (116)$$

Hence

$$\boxed{\partial_\mu J^\mu = 0.} \quad (117)$$

Expanding,

$$\partial_\mu J^\mu = \partial_0 J^0 + \partial_i J^i \quad (118)$$

$$= \frac{1}{c} \frac{\partial}{\partial t} (c\rho) + \nabla \cdot \mathbf{J}, \quad (119)$$

so

$$\boxed{\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0.} \quad (120)$$

Thus local charge conservation is naturally encoded in the covariant structure of Maxwell's equations.

14 Gauge freedom

Using

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu,$$

consider

$$A_\mu \longrightarrow A'_\mu = A_\mu + \partial_\mu \Lambda, \quad (121)$$

where $\Lambda(x)$ is a scalar function.

Then

$$F'_{\mu\nu} = \partial_\mu A'_\nu - \partial_\nu A'_\mu \quad (122)$$

$$= F_{\mu\nu} + \partial_\mu \partial_\nu \Lambda - \partial_\nu \partial_\mu \Lambda \quad (123)$$

$$= F_{\mu\nu}. \quad (124)$$

Therefore

$$\boxed{F'_{\mu\nu} = F_{\mu\nu}.} \quad (125)$$

This is the gauge invariance of classical electromagnetism.

15 Maxwell's equations in terms of the four-potential

Substituting

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$$

into

$$\partial_\mu F^{\mu\nu} = -\mu_0 J^\nu,$$

we get

$$\partial_\mu (\partial^\mu A^\nu - \partial^\nu A^\mu) = -\mu_0 J^\nu. \quad (126)$$

Thus

$$\square A^\nu - \partial^\nu (\partial_\mu A^\mu) = -\mu_0 J^\nu, \quad (127)$$

where

$$\square \equiv \partial_\mu \partial^\mu = -\frac{1}{c^2} \frac{\partial^2}{\partial t^2} + \nabla^2. \quad (128)$$

In Lorenz gauge,

$$\boxed{\partial_\mu A^\mu = 0,} \quad (129)$$

the field equation becomes

$$\boxed{\square A^\nu = -\mu_0 J^\nu.} \quad (130)$$

In vacuum,

$$J^\nu = 0,$$

so

$$\boxed{\square A^\nu = 0.} \quad (131)$$

16 Logical summary

The entire construction can be organized as the following chain of reasoning:

1. Special relativity: fundamental laws should be written in a form preserved by Lorentz transformations.

2. Charge and current: the proper charge density multiplied by the four-velocity gives

$$J^\mu = \rho_0 U^\mu = (c\rho, \mathbf{J}).$$

Therefore charge density and current density are components of a single four-vector.

3. Relativistic force law: the Lorentz force must become a four-vector equation. Assuming linear dependence on U^μ ,

$$f^\mu = q F^\mu{}_\nu U^\nu.$$

4. Antisymmetry: fixed rest mass implies

$$U_\mu f^\mu = 0,$$

which forces

$$F_{\mu\nu} = -F_{\nu\mu}.$$

5. Identification of the field components: comparison with

$$\mathbf{f} = q(\mathbf{E} + \mathbf{u} \times \mathbf{B})$$

gives

$$F^{0i} = \frac{E_i}{c}, \quad F^{ij} = \varepsilon_{ijk} B_k.$$

6. Lorentz transformations: the tensor rule

$$F'^{\mu\nu} = \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta F^{\alpha\beta}$$

implies the observed mixing of electric and magnetic fields.

7. Homogeneous Maxwell equations:

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = 0$$

become

$$\partial_{[\alpha} F_{\beta\gamma]} = 0.$$

8. Inhomogeneous Maxwell equations:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}, \quad \nabla \times \mathbf{B} - \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t} = \mu_0 \mathbf{J}$$

become

$$\partial_\mu F^{\mu\nu} = -\mu_0 J^\nu.$$

9. Charge conservation: antisymmetry of $F^{\mu\nu}$ implies

$$\partial_\mu J^\mu = 0.$$

The central conceptual message is therefore:

$$\boxed{(\rho, \mathbf{J}) \text{ are components of } J^\mu, \quad (\mathbf{E}, \mathbf{B}) \text{ are components of } F_{\mu\nu}.} \quad (132)$$

The relativistic formulation does not merely rewrite Maxwell's equations more compactly. It reveals the geometrical structure that is hidden in their ordinary three-dimensional form.